

The Event-State Universe: A Computational Synthesis

Torben Wille
Independent Researcher
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Abstract

We present a resolution to long-standing cosmological tensions by synthesizing a formal discrete causal framework with large-scale numerical simulation. The Event-State Theory (EST) posits a universe evolving through discrete global states selected by a cost-minimization principle. A 3D computational kernel implementing this framework spontaneously generates a cosmic web, a linear arrow of time, and a matter-dominated universe from initial random conditions. The simulation output resolves the Hubble Tension as a natural consequence of asynchronous causal structure formation and predicts a novel pre-causal observational signature. The complete source code and data are provided for verification.

I. INTRODUCTION

The Hubble Tension [1] and the baryon asymmetry problem represent fundamental challenges to the standard cosmological model. Proposals within the continuum paradigm have failed to provide a compelling, unified solution. We propose that these are not independent puzzles but symptoms of a single underlying cause: the assumption of continuous spacetime.

In a companion paper [2], we introduced the axioms of Event-State Theory (EST), a discrete causal-informational framework. In a second, results-oriented paper [3], we demonstrated that a computational kernel based on these axioms produces a universe with striking morphological similarity to our own. This work synthesizes these two strands, presenting EST as a complete, empirically-grounded framework and deriving its consequences for cosmology.

II. THE EST FRAMEWORK IN BRIEF

EST is built on three axioms:

1. **Discrete Causal Frames:** Reality is a sequence of global states Ψ_n .
2. **Topological Constant:** $c \equiv 1$ (node/frame). The speed of light is a connectivity constraint.
3. **Computational Least Action:** $P(\Psi_{n+1}) \propto \exp(-J/\lambda)$, where the cost function J balances energy change ΔE against algorithmic complexity K .

III. FROM AXIOMS TO A SIMULATED COSMOS

The kernel described in [3] operationalizes these axioms. It evolves a 3D lattice by probabilistically selecting subsequent states that minimize $J = \alpha\Delta E + \beta K + \epsilon$, where ϵ is a small matter-asymmetry bias. The parameters $\beta \approx 3.4$, $\lambda \approx 0.48$, $\epsilon \approx 0.075$ were found to be an attractor producing stable, structured universes.

As shown in Fig. 1 of [3], this kernel spontaneously generates a cosmic web. Figure 2 of the same work demonstrates a monotonic increase in matter density, providing a mechanistic account of baryogenesis without fine-tuning.

IV. RESOLVING THE HUBBLE TENSION

The EST framework naturally gives rise to the Asynchronous Multi-Point Crystallization (AMPC) model. The primordial universe was not a single, synchronous explosion but a cascade of local crystallizations in the field of potential. The observed local Hubble parameter H_0 is a superposition of contributions from these nucleation centers:

$$H(x) \approx \sum_i \frac{A_i}{\|x - C_i\|},$$

where A_i is the Causal Propagation Rate. Regions of low matter density (voids) present less computational resistance, leading to a higher local A_i and a faster observed expansion rate ($H_0 \approx 73$). Dense regions (filaments) have lower A_i and a slower rate ($H_0 \approx 67$). The tension is not an error but a measurement of the underlying computational geometry.

V. PREDICTION: PRE-CAUSAL RESONANCE

A fundamental consequence of the EST dynamical mechanism is that the crystallization of an event-state involves the entire local field of potential Ω . This generates a testable prediction: correlated vacuum fluctuations, or **Pre-Causal Resonance (PCR)**, should be detectable prior to the nominal $t = 0$ of a high-energy collision event. A conclusive null result for PCR falsifies the core state-selection mechanism of EST.

VI. CONCLUSION

The synthesis of the formal EST framework with large-scale computational simulation presents a formidable challenge to continuum-based cosmology. EST resolves the Hubble Tension and baryon asymmetry from first principles, provides a mechanism for large-scale structure formation, and makes a novel, falsifiable prediction. The code and data are publicly available [4] for independent verification. The evidence compels a serious reevaluation of spacetime's fundamental nature.

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- [1] E. Di Valentino *et al.*, In the realm of the hubble tension—a review of solutions, *Classical and Quantum Gravity* **38**, 153001 (2021).
 - [2] T. Wille, Event-state theory: A discrete causal-information framework for cosmology and fundamental physics (2024).
 - [3] T. Wille, Computational emergence of cosmic structure from algorithmic probability (2025).
 - [4] T. Wille, Event-state protocols github repository, <https://github.com/SirWillance/Event-State-Theory-EST-> (2025).